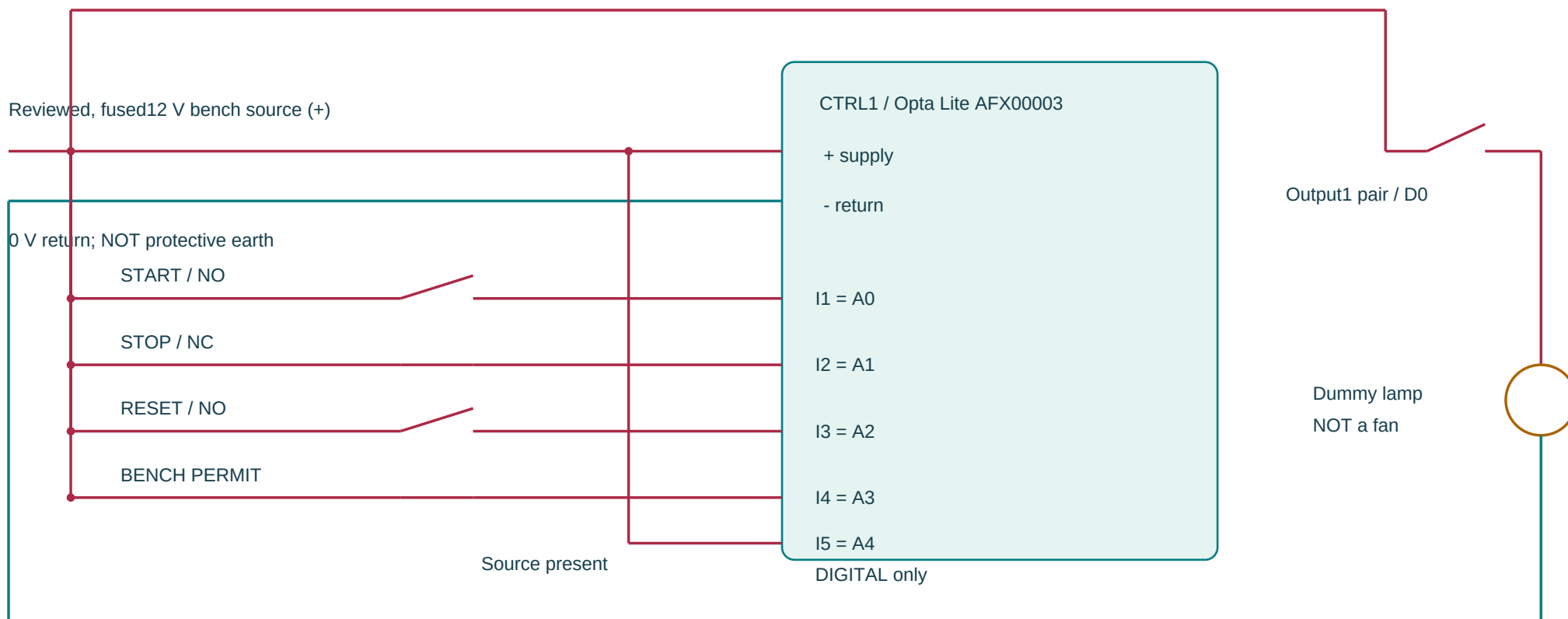


A real controller and implemented local sequence

Terminal reference for a protected dummy-load bench only. PR1 and the tower protective system remain OPEN.



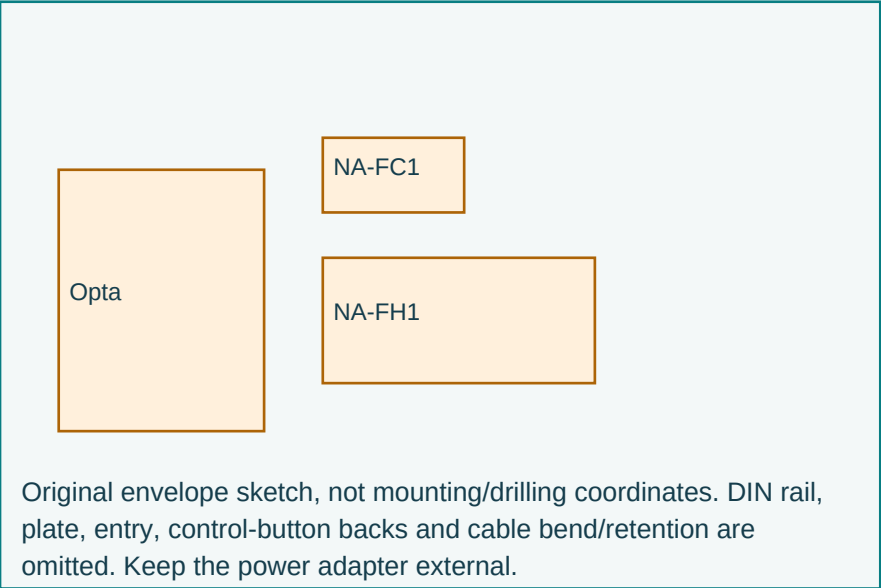
Connection hold: junction dots connect; crossings without dots do not. Contacts and power protection are not approved parts. Output1 is its printed contact pair, not invented COM/NO identifiers. Preserve polarity; compatible protected harness required. No connection to the tower fan chain is released.

Firmware: all relay commands remain LOW by default. RUN REQUESTED/TRIP/source LEDs show software status only. Inputs I1–I5 are digital. 12 V is outside the 0–10 V analog range; do not change I5 to analog mode. I4 is a bench test signal, not independent protective permission.

Enclosure fit and protection coordination

A concrete external-box candidate; not a fabrication layout, released fuse choice or completed safety circuit.

Hammond1554XA2GY /300 x200 x120 mm



Fit finding: Opta front70 x88.8 mm; drawing-derived depth reservation61.1 mm. A40 mm deep CAD box is not a normal DIN fit. Do not reinterpret the stock box IP rating as modified-device weather protection. The new external box is a recommendation; native tower CAD is unchanged.

Power: Opta maximum2 W at12 V adds0.167 A. Fans plus Opta alone total1.567 A /18.8 W. Supply margin0.433 A remains before hub/controller, input circuits and startup. No fan power-budget approval follows.

Real fuse screen	Verified OEM data	Decision / consequence
Littelfuse0297002.WXNV	2 A /32 V DC;1000 A interrupt rating at32 V	SCREENED, not selected. Holder/actual circuit coordination unresolved
Typical thermal allowance	1.9 A at20 C;1.7 A at40 C	Fans+Opta leave only0.133 A to40 C allowance before other loads
Opening at110% /2.2 A	Minimum360000 s; no maximum given	Not a2 A current limiter or proof of prompt clearing on a limited supply
Opening at200% /4 A	0.15–5 s specified window	Actual PSU/hub fault current/time and weakest cable/contact required

Limits: 1,041 host assertions pass, including1,024 one-step cases. Board compilation and all physical tests are NOT DONE. Software, ordinary relays, digital source presence and LEDs do not prove safe access, fault coverage or standstill. Existing restart rules remain unchanged and unresolved for internal fan/hub resets.